JEE Mains 2020 Jan Chapter wise Question Bank

Complex Numbers

Q1

If z = x + iy and real part $\left(\frac{z-1}{2z+i}\right) = 1$ then locus of z is

(1) Straight line with slope 2

(2) Straight line with slope $-\frac{1}{2}$

(3) circle with diameter $\frac{\sqrt{5}}{2}$

(4) circle with diameter $\frac{1}{2}$

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Sol

(3)

z = x + iy

$$\left(\frac{z-1}{2z+i}\right) = \frac{(x-1)+iy}{2(x+iy)+i} = \frac{(x-1)+iy}{2x+(2y+1)i} \times \frac{2x-(2y+1)i}{2x-(2y+1)i}$$

$$\text{Re}\left(\frac{z+1}{2z+i}\right) = \frac{2x(x-1)+y(2y+1)}{(2x)^2+(2y+1)^2} = 1$$

02

If $z = \left(\frac{3+i\sin\theta}{4-i\cos\theta}\right)$ is purely real and $\theta \in \left(\frac{\pi}{2}, \pi\right)$ then arg(sinθ + i cosθ) is -

यदि $z = \left(\frac{3+i\sin\theta}{4-i\cos\theta}\right)$ पूर्णतया वास्तविक है तथा $\theta \in \left(\frac{\pi}{2}, \pi\right)$ तो arg(sinθ + i cosθ) का मान होगा–

(1) $-\tan^{-1}\frac{3}{4}$ (2) $\pi - \tan^{-1}\frac{3}{4}$ (3) $\pi - \tan^{-1}\frac{4}{3}$ (4) $\tan^{-1}\frac{4}{3}$

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Sol

(3)

$$z = \frac{(3+i\sin\theta)}{(4-i\cos\theta)} \times \frac{(4+i\cos\theta)}{(4+i\cos\theta)}$$

as चूंकि z is purely real पूर्णतः वास्तविक है $\Rightarrow 3\cos\theta + 4\sin\theta = 0 \Rightarrow \tan\theta = -\frac{3}{4}$

$$\arg(\sin\theta + i\cos\theta) = \pi + \tan^{-1}\left(\frac{\cos\theta}{\sin\theta}\right) = \pi + \tan^{-1}\left(-\frac{4}{3}\right) = \pi - \tan^{-1}\left(-\frac{4}{3}\right)$$

03

Roots of the equation $x^2 + bx + 45 = 0$, $b \in R$ lie on the curve $|z + 1| = 2\sqrt{10}$, where z is a complex number then

समीकरण $x^2 + bx + 45 = 0$, $b \in R$ के मूल वक्र $|z + 1| = 2\sqrt{10}$ पर स्थित है, जहाँ z सम्मिश्र संख्या है तब

(1) $b^2 + b = 12$ (2) $b^2 - b = 30$ (3) $b^2 - b = 36$ (4) $b^2 + b = 30$

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Sol

(2)

Let $z = \alpha \pm i\beta$ be roots of the equation

माना $z = \alpha \pm i\beta$ समीकरण के मूल है

So $2\alpha = -b$ and $\alpha^2 + \beta^2 = 45$, $(\alpha + 1)^2 + \beta^2 = 40$

इसलिए $2\alpha = -b$ तथा $\alpha^2 + \beta^2 = 45$, $(\alpha + 1)^2 + \beta^2 = 40$

So $(\alpha + 1)^2 - \alpha^2 = -5$

इसलिए $(\alpha + 1)^2 - \alpha^2 = -5$

$\Rightarrow \quad 2\alpha + 1 = -5 \quad \Rightarrow \quad 2\alpha = -6$

so b = 6

इसलिए b = 6

hence $b^2 - b = 30$ अत: $b^2 - b = 30$

04

Let $\alpha = \frac{-1+i\sqrt{3}}{2}$ and $a = (1+\alpha)\sum_{k=0}^{100} \alpha^{2k}$, $b = \sum_{k=0}^{100} \alpha^{3k}$. If a and b are roots of quadratic equation then quadratic equation is

माना कि $\alpha = \frac{-1+i\sqrt{3}}{2}$ तथा $a = (1+\alpha)\sum_{k=0}^{100} \alpha^{2k}$, $b = \sum_{k=0}^{100} \alpha^{3k}$ यदि a तथा b द्विघात समीकरण के मूल है तो द्विघात समीकरण है–

(1) $x^2 - 102x + 101 = 0$ (2) $x^2 - 101x + 100 = 0$

(3) $x^2 + 101x + 100 = 0$ (4) $x^2 + 102x + 100 = 0$

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Sol

(1)

$\alpha = \omega$, $b = 1 + \omega^3 + \omega^6 + \ldots\ldots = 101$

$a = (1 + \omega)(1 + \omega^2 + \omega^4 + \ldots..\omega^{198} + \omega^{200})$

$$= (1+\omega)\frac{\left(1-\left(\omega^2\right)^{101}\right)}{1-\omega^2} = \frac{(1+\omega)(1-\omega)}{1-\omega^2} = 1$$

Equation : समीकरण $x^2 - (101 + 1)x + (101) \times 1 = 0$ $\Rightarrow x^2 - 102x + 101 = 0$

05

If $\left|\frac{z-i}{z+2i}\right| = 1$, $|z| = \frac{5}{2}$ then value of $|z + 3i|$ is

यदि $\left|\frac{z-i}{z+2i}\right| = 1$, $|z| = \frac{5}{2}$ तो $|z + 3i|$ का मान है–

(1) $\frac{7}{2}$ (2) $\sqrt{10}$ (3) $\sqrt{5}$ (4) $\sqrt{3}$

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Sol

(1)

$x^2 + (y-1)^2 = x^2 + (y+2)^2$

$-2y + 1 = 4y + 4$

$6y = -3 \Rightarrow y = -\frac{1}{2}$

$x^2 + y^2 = \frac{25}{4} \Rightarrow x^2 = \frac{24}{4} = 6$

$\Rightarrow \quad z = \pm\sqrt{6} - \frac{i}{2}$

$|z + 3i| = \sqrt{6 + \frac{25}{4}} = \sqrt{\frac{49}{4}}$

$|z + 3i| = \frac{7}{2}$

06

z is a complex number such that $|Re(z)| + |Im(z)| = 4$ then $|z|$ can't be

z एक समिश्र संख्या इस प्रकार है कि $|Re(z)| + |Im(z)| = 4$ तब $|z|$ नही हो सकता

(1) $\sqrt{7}$ (2) $\sqrt{10}$ (3) $\sqrt{\frac{17}{2}}$ (4) $\sqrt{8}$

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Sol

(1)

$z = x + iy$

$|x| + |y| = 4$

Minimum value of $|z| = 2\sqrt{2}$

Maximum value of $|z| = 4$

$|z| \in \left[\sqrt{8}, \sqrt{16}\right]$

So $|z|$ can't be $\sqrt{7}$